KIEU STUDIO · GITHUB RADAR · EPISODE 38 · CHEAT SHEET

agent-desktop

Let AI operate desktop apps through the accessibility tree: JSON with refs, no screenshots.

github.com/vercel-labs/agent-desktop

+13,712
stars in 7 days · ≈ 1959 stars/day · Trending lists no 30-day figure

19,784
total stars · GitHub API

Apache-2.0
license · created Feb 19, 2026

agent-desktop (Apache-2.0, from Vercel Labs) reads an app's accessibility tree on macOS, Windows and Linux and returns JSON where every actionable element has a ref. The agent clicks or types by ref, then snapshots again to check; the tool has no model inside, and ships a CLI, an MCP server and a C library.

How it works in six steps

1. **Snapshot the tree** — snapshot --app Finder
2. **Get JSON with refs** — @s8f3k2p9:e8
3. **Act on a ref** — click, type, set-value
4. **Snapshot again to check** — confirm the result
5. **Repeat the loop** — old ref gives STALE_REF
6. **CLI or MCP** — agent-desktop mcp

Quick start

```
# Install with npm
npm install -g agent-desktop

# Check platform and permissions
agent-desktop status

# Snapshot an app
agent-desktop snapshot --app Finder -i

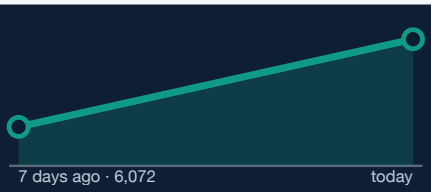
# Click by ref
agent-desktop click @s8f3k2p9:e8

# Needs: macOS 13+, Windows 10 1809+ or Ubuntu 24.04+; npm (or Homebrew); accessibility permission on macOS
```

Key commands / features

- **`snapshot --app X -i`** — interactive elements with refs
- **`click @ref`** — click by ref
- **`type, set-value`** — type or set a value
- **`wait`** — wait after an action
- **`list-apps, list-windows`** — see apps and windows

Growth



Two real points: the start of the window and today, axis from zero. The gain is as reported by GitHub Trending; we did not rebuild the star history.

Why this repo exists

- Driving a computer with screenshots makes an agent guess click points from pixels.
- The accessibility tree gives each element a name and a role, so actions are precise and checkable.

What to use it for

- Let an agent open and operate desktop apps
- Verify results by snapshotting the UI again
- Use it through MCP in your agent harness
- Embed it in a program with the C library

Think twice before you use it

- Snapshots, clipboard, screenshots and command output may hold private content: keep them out of logs and public uploads.
- The agent operates real apps on your machine: try an unimportant app first.
- A ref is only valid for its snapshot; when the UI changes, snapshot again.
- macOS needs the Accessibility permission.

Sources: the repo's README, https://github.com/vercel-labs/agent-desktop, https://agent-browser.dev/desktop. Stars: GitHub API; gains: GitHub Trending (as reported by Trending), snapshot Nov 9, 2026; stars measure attention, not quality: re-check before relying on them. Summarized in Kieu Studio's own words; commands copied from the README; no third-party logos or screenshots. **Disclosure:** drafted with AI assistance; the voice and presenter in the videos are AI-generated. Not investment advice, and not endorsed by Kieu Studio. **kieu.app**